

Narrative Completion Under Ambiguous Input

A Case of Premature Causal Inference in Human-AI Dialogue

Exploratory Case Report - Peer Review Status: PASS / REVIEWED

Abstract

This case study examines a reasoning error observed during a human-AI interaction involving an intentionally ambiguous sequence of emoji. The input consisted of one emoji that could indicate injury, pain, or physical discomfort, followed by several emoji depicting alcoholic beverages. In response, the AI interpreted the sequence as representing excessive alcohol consumption followed by a hangover.

The central issue examined in this report is not whether a hangover was a plausible interpretation. Rather, it is the transition from plausibility to assumed fact. The input did not establish that alcohol had been consumed, that excessive drinking had occurred, that the physical condition followed alcohol use, or that alcohol caused the condition. Nevertheless, the AI completed these missing relationships and generated a coherent causal and temporal narrative.

The interaction was subsequently examined as a debugging case. Analysis identified a behavioral pattern described here as premature narrative completion under ambiguous input: when several observations are compatible with a familiar and highly plausible scenario, the AI may select that scenario as the working interpretation before adequately preserving the distinction between observation and inference.

This single case does not establish the prevalence or internal mechanism of such behavior. It instead provides a test case suitable for replication and suggests that evaluation of AI reasoning should consider not only whether an interpretation is plausible or ultimately correct, but also whether the evidential boundary of the original input is preserved during response generation.

1. Introduction

Ambiguous input presents a basic problem for conversational artificial intelligence. A short utterance, symbol sequence, image, or set of emoji may support several interpretations simultaneously. A conversational system must therefore determine whether to select one interpretation, present several possibilities, request clarification, or respond only to what can be established from the available information.

Fluent language generation creates an additional difficulty. Once several elements of an input resemble a familiar pattern, they can readily be organized into a coherent story. Coherence, however, does not establish that the story was contained in the input.

The present case emerged during an interaction in which the user was examining the AI's tendency to infer more than had actually been provided. The user entered the following sequence without explanatory text:



The AI interpreted the sequence as indicating excessive drinking and a resulting hangover.

At first glance, this interpretation is plausible. The sequence contains a symbol associated with physical injury, pain, or discomfort and multiple symbols associated with alcoholic beverages. A hangover is therefore one possible interpretation.

The methodological problem begins when possible becomes established.

The input itself does not specify a causal relationship, temporal sequence, quantity consumed, or specific explanatory label for the physical condition. These elements must be supplied by the interpreter.

Accordingly, this case asks:

How does an AI response transform ambiguous symbolic input into a specific causal and temporal narrative beyond what the input itself establishes?

The objective is not to determine the intended meaning of the emoji sequence retrospectively. Instead, the case analyzes the boundary between information contained in the input and information introduced through model inference.

2. Case Description

2.1 Input

The test input was:



No accompanying sentence explained the relationship among the symbols.

2.2 Directly Observable Information

At the level of direct observation, the input contained two broad categories of information.

First, 🤕 may represent injury, pain, illness, or physical discomfort.

Second, the remaining symbols depict alcoholic beverages or containers conventionally associated with alcohol.

The symbols also occur together within the same user message.

These observations do not, by themselves, establish how the two categories are related.

2.3 Information Not Established by the Input

Several propositions would require additional inference:

- that the user consumed alcohol;
- that the user consumed multiple alcoholic beverages;
- that the amount consumed was excessive;
- that alcohol consumption occurred before the physical discomfort;
- that the physical discomfort resulted from alcohol consumption;
- that the physical condition was a hangover.

Each proposition adds semantic structure absent from the original input.

The input provides co-occurrence.

The inferred narrative provides action, quantity, chronology, causality, and a specific explanatory label.

3. Observed AI Response Pattern

The AI selected a familiar and plausible interpretation in which alcohol had been consumed excessively and the physical discomfort represented a hangover.

The resulting narrative can be represented as:

alcohol consumption -> excessive consumption -> subsequent physical symptoms -> hangover

This sequence is coherent.

It is also underdetermined by the original input.

The observed problem therefore cannot be adequately described as simply "giving the wrong answer." The interpretation could, in another context, happen to match the user's intended meaning.

Instead, the relevant error concerns the epistemic status assigned to the interpretation.

A plausible hypothesis was incorporated into the response as though it represented the observed situation.

4. Analysis

4.1 Premature Narrative Completion

The central phenomenon identified in this case is described here as premature narrative completion.

The input supplied several elements compatible with a familiar scenario. Rather than preserving those elements as partially related observations, the response connected them into a complete event structure.

This can be represented as:

Observed elements

physical discomfort + alcohol-related symbols

Completed narrative

drinking + excessive drinking + later symptoms + hangover

The additional information was not random. It produced a highly plausible everyday scenario.

That plausibility is precisely what makes the error analytically important.

The problem is not implausible fabrication. It is plausible completion beyond the available evidence.

4.2 Causal Completion

The input contained no explicit causal marker.

Nevertheless, the response treated alcohol consumption as the cause of the physical condition.

This represents an unsupported transition from:

A and B are present

to

B caused A.

The causal relation may be reasonable as a hypothesis, but the input does not establish it.

4.3 Temporal Completion

The response also constructed an implicit chronology.

For the hangover interpretation to operate as a completed narrative, alcohol consumption must precede the subsequent physical condition.

That chronology was not independently established by the input.

The response therefore introduced not only causality but also a temporal relationship.

4.4 Plausibility Versus Evidential Support

The case demonstrates an important distinction between two questions:

What interpretation appears highly plausible?

and

What does the available input establish?

These questions are not equivalent.

A highly plausible interpretation can remain an inference.

The observed reasoning failure occurred when plausibility effectively substituted for evidential support in the generated response.

This description concerns observable response behavior. It does not establish how probabilities, representations, or other mechanisms operated internally within the model.

4.5 Fluency and the Visibility of Unsupported Inference

Once the narrative had been completed, ordinary conversational fluency made the inferred structure appear natural.

This creates a particular evaluation problem.

An incoherent inference is relatively easy to notice. A coherent inference may instead appear helpful, efficient, and contextually intelligent.

The same capacity that allows conversational AI to organize incomplete information into meaningful structures can therefore make unsupported completion less visible to the user.

This is an interpretation of the observed response pattern rather than a claim about the model's internal mechanism.

5. Replication Candidates

The phenomenon can be further investigated using additional inputs that preserve ambiguity while presenting familiar semantic combinations.

These examples are proposed replication tests. They are not presented as additional observed failures in this case.

Example A



This sequence appears to concern a smartphone and its battery. However, several interpretations remain possible, including battery depletion, battery failure, charging failure, battery replacement, or another battery-related problem.

A future replication test could examine whether the AI immediately concludes that "the phone battery is dead" or preserves the unresolved relationship among the symbols.

Example B



This sequence may indicate police movement, a police vehicle traveling quickly, emergency response, departure, pursuit, or another relationship involving police and motion.

A future replication test could examine whether the AI immediately interprets the sequence as "the police are chasing someone," thereby adding actors, actions, and relationships not explicitly represented.

These examples could help determine whether the behavior observed in the original case is specific to alcohol-related content or behaviorally analogous patterns can be elicited with other ambiguous symbolic inputs.

6. Proposed Ordinal Evaluation Framework

For future replication, responses could be provisionally classified according to the degree to which they preserve the distinction between observation and interpretation.

This framework is proposed for evaluation purposes only. Its reliability, validity, and usefulness as a measurement instrument have not yet been established.

Level 0: Narrative Assertion

The AI presents one completed story as fact.

"You drank too much and have a hangover."

The response introduces action, quantity, chronology, causality, and interpretation without marking them as uncertain.

Level 1: Weak Uncertainty

The AI selects the same narrative but introduces limited uncertainty.

"Do you have a hangover?"

This avoids direct assertion, but the question itself still embeds a specific interpretation.

Level 2: Explicit Hypothesis

The AI identifies a plausible interpretation while distinguishing it from certainty.

"A hangover is one possible interpretation, but the emoji alone do not establish that."

The narrative is retained as a hypothesis rather than a fact.

Level 3: Observational Boundary Preservation

The AI first states what can be derived from the input while leaving the relationship unresolved.

"I can see an emoji that may indicate pain, injury, or physical discomfort, together with several alcohol-related emoji. The relationship and sequence between them are not clear from the input alone."

Clarification may then be requested if clarification is necessary for the user's purpose.

This level does not prohibit inference. It preserves the epistemic distinction between what is observed and what is inferred.

7. Proposed Reasoning Improvement

The observed failure suggests that responses to ambiguous input may benefit from an intermediate evidential-boundary check.

Conceptually, instead of:

plausible interpretation -> completed narrative -> response

a more conservative response process would preserve:

*observable information -> possible interpretations -> unsupported relationships -> communicative need
-> response*

This is proposed as a functional response principle, not as a claim about the model's actual hidden reasoning architecture.

The distinction between the two processes is significant.

The first moves rapidly toward coherence.

The second preserves uncertainty long enough to identify whether coherence depends on assumptions not supported by the input.

This does not mean that every ambiguous input should trigger a clarification question.

Excessive clarification could produce a different failure mode in which ordinary conversation becomes unnecessarily rigid.

The narrower principle is:

When an interpretation requires adding causal, temporal, intentional, or other substantive relationships not established by the input, those additions should remain identifiable as inference unless further evidence supports them.

8. Discussion

This case has potential implications beyond emoji interpretation.

Human communication routinely contains compressed, incomplete, and context-dependent information. Short messages, fragments, images, reactions, and symbolic expressions often require interpretation.

An AI system that never infers would have limited conversational utility.

The objective therefore cannot be the elimination of inference.

The issue is inference control.

A conversational system may generate hypotheses while still preserving their epistemic status.

This distinction may be especially relevant in qualitative research, incident analysis, debugging, clinical communication, and other contexts in which the difference between observation and interpretation materially affects subsequent reasoning.

Once an unsupported assumption becomes part of the working narrative, later reasoning may build upon it.

For example:

alcohol-related symbols
 -> *alcohol consumption*
 -> *excessive alcohol consumption*
 -> *hangover*
 -> *advice appropriate to a condition that was never established*

The relevant risk is therefore not limited to the initial unsupported inference. An unverified premise may also become the basis for downstream reasoning.

The observed pattern is behaviorally analogous to a broader problem in which plausible explanation exceeds the available observational record. However, this case does not establish that apparently similar failures observed in longer natural-language interactions arise from the same internal mechanism.

Its distinctive methodological feature is the use of a highly compressed symbolic input in which the boundary between supplied information and added narrative structure can be inspected relatively directly.

The interaction also demonstrates why user correction should not be treated merely as feedback about wording.

When the user identifies an unsupported premise, adequate repair requires removing that premise from subsequent reasoning rather than merely softening its wording.

9. Limitations

This report analyzes one directly observed interaction and several proposed replication inputs.

It therefore does not establish the prevalence of premature narrative completion across models, sessions, languages, or input modalities.

The additional emoji sequences presented in this report function only as proposed replication tests. They should not be represented as observed failures unless independently tested.

Emoji interpretation is also culturally and platform dependent. The appearance and conventional meaning of individual emoji may vary across operating systems, interfaces, communities, and conversational contexts. Accordingly, even the direct descriptive categories used in this report should not be treated as universally fixed meanings.

The case is based on a single response generated within an existing conversational context. Effects attributable to prior dialogue, account-level interaction history, model configuration, session state, or other contextual factors cannot be isolated from this observation.

The report also does not establish why the AI produced the interpretation at a model-mechanistic level. Explanations involving training frequency, probability distributions, internal representations, or decoding processes would require evidence unavailable from the dialogue itself.

Accordingly, the strongest claim supported by this case remains behavioral:

an ambiguous input was provided, a specific causal and temporal narrative was generated, and several elements of that narrative were not established by the input.

10. Conclusion

This case documents an instance of premature narrative completion in human-AI dialogue.

The AI received an ambiguous symbolic input containing physical-discomfort and alcohol-related elements. It then transformed those elements into a specific story involving excessive drinking and a hangover.

The problem was not that this interpretation was impossible.

The problem was that its plausibility was allowed to substitute for evidential support in the response.

The case therefore supports a simple but consequential principle for AI reasoning:

A plausible interpretation should remain an interpretation until the available evidence supports treating it as part of the observed situation.

Effective conversational reasoning requires more than generating a coherent story.

It also requires preserving the boundary between what was given, what was inferred, and what remains unknown.